



...**LIGHTCITY**...

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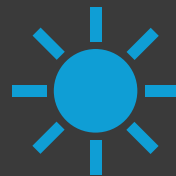
Problems



No communication between lights



Fixing broken lights



Wasting energy

Scenarios

Happy flow:

- Single Pedestrian
- Two Pedestrian walking in opposite direction

Unhappy flow:

- Network Crash
- Broken Light/Node

Bordercase:

- In between the lights
- Enter at a broken node



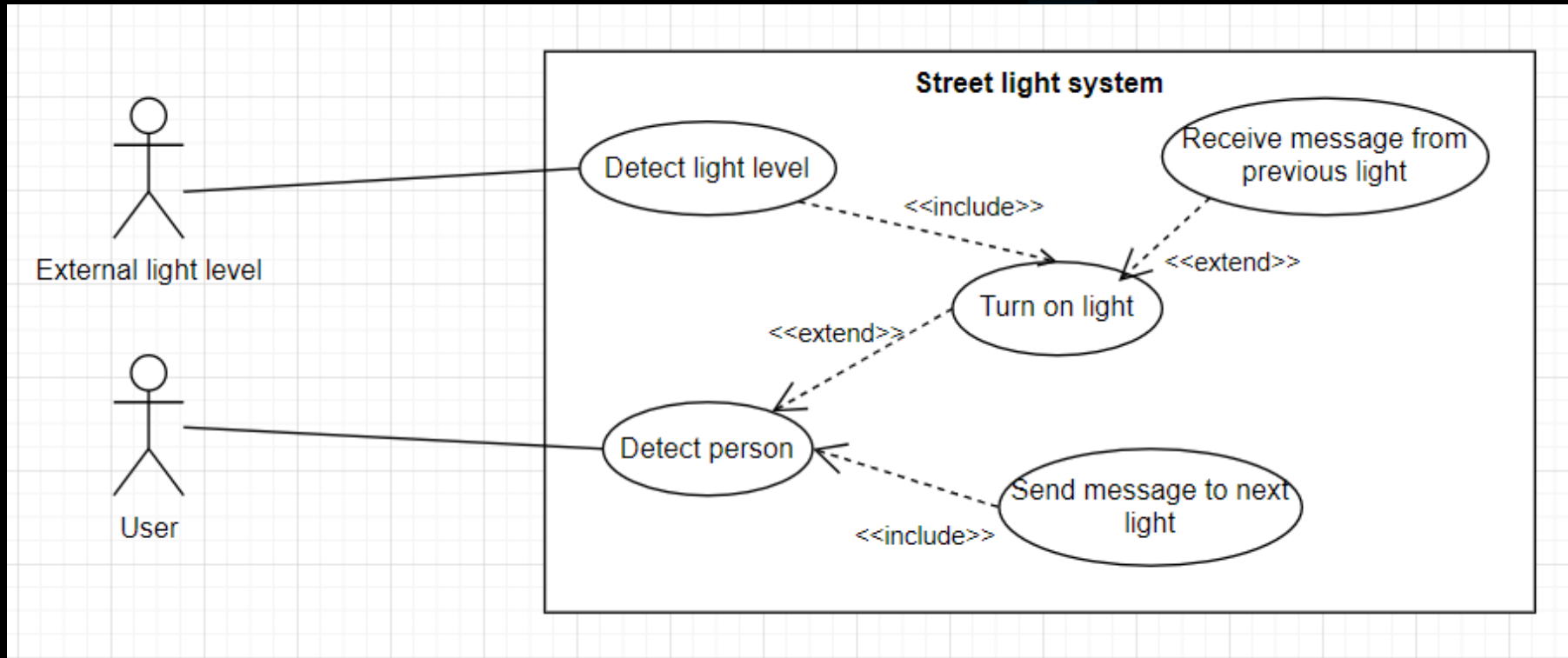
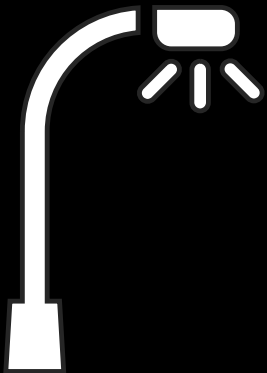
MUST Requirements

- Faulty node
- The light turns on full intensity when triggered
- The lights gradually dims
- Maximum brightness threshold of LDR -> lights off.
- Minimum brightness threshold of LDR -> minimum brightness.

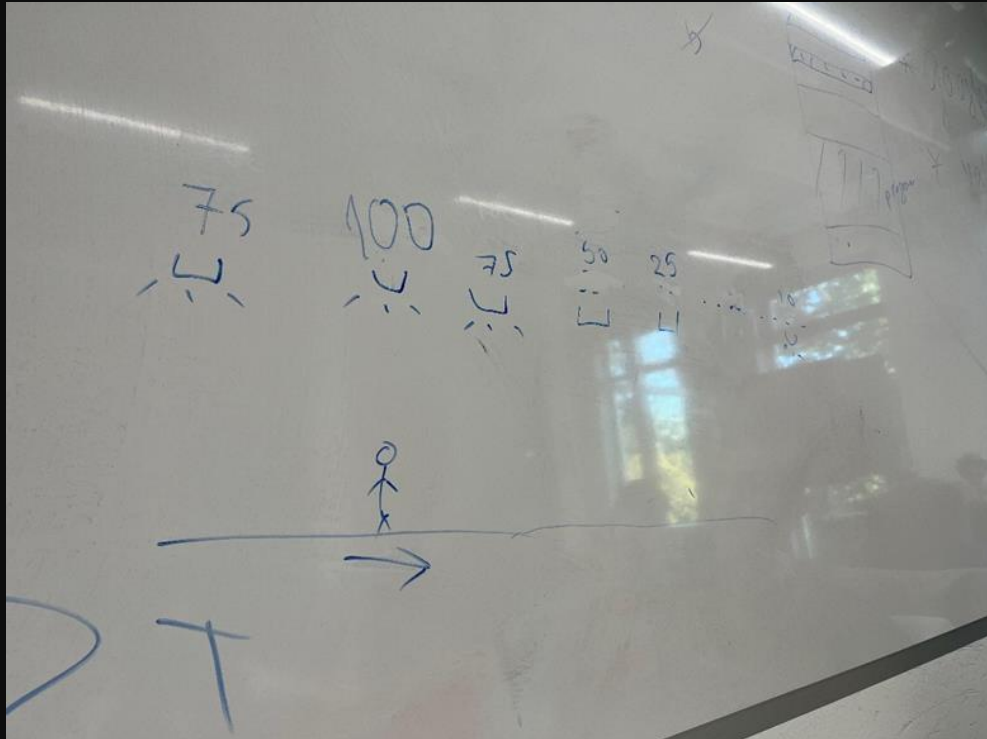


Use Cases

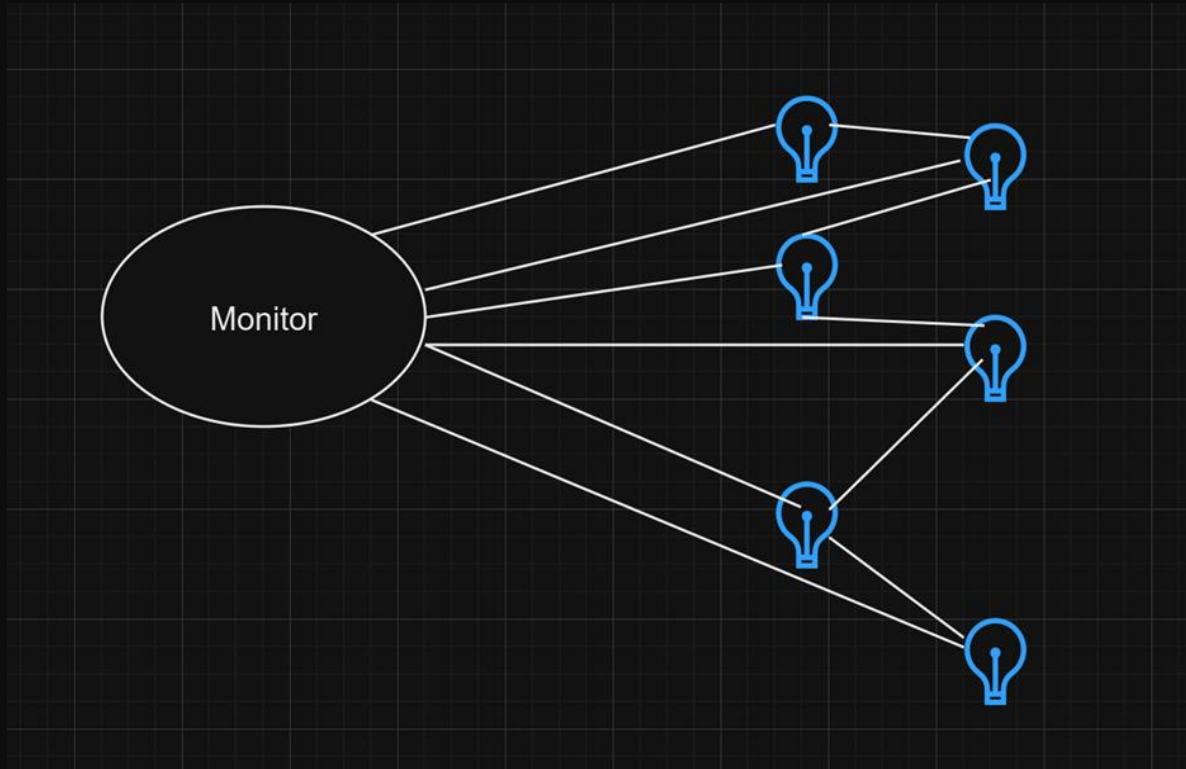
- Adjust brightness
- Adjust lights to environment
- Turn on lights during network outage



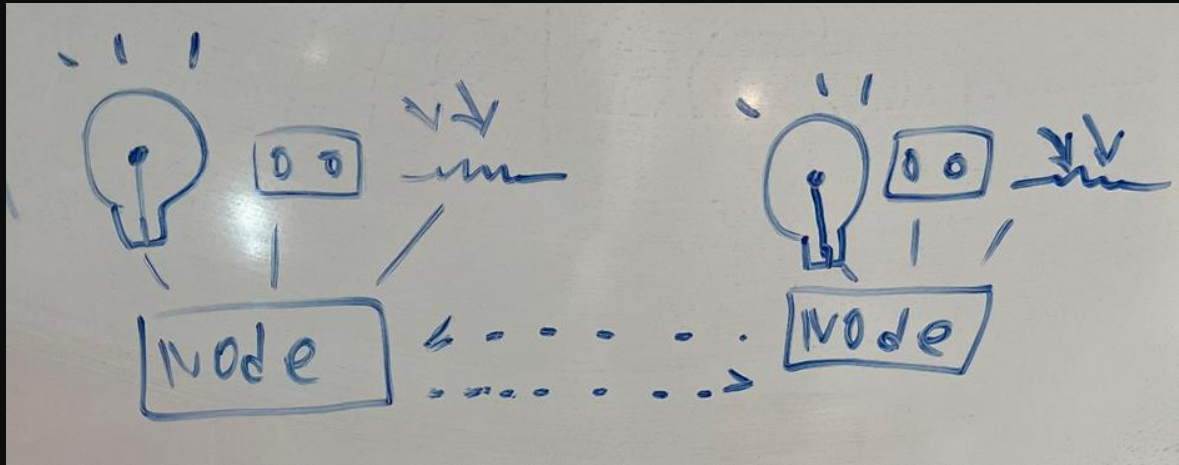
Environment Interaction



System Structure



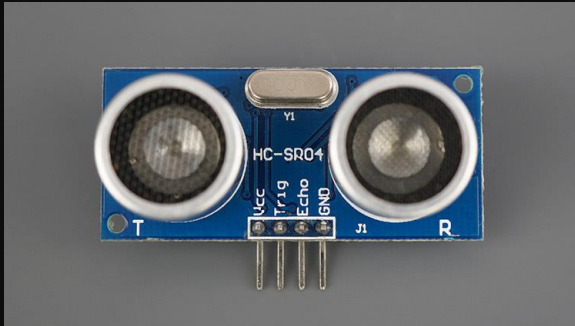
System Interactions



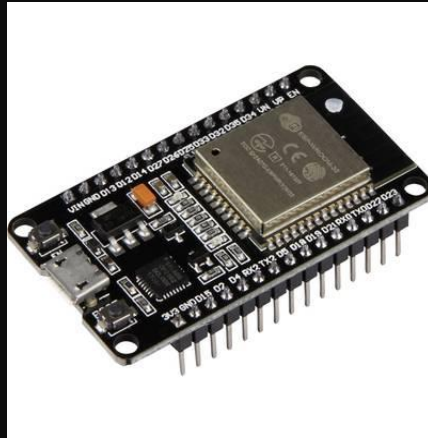
Message Protocol

MESSAGE	INTERVAL	WHEN
Motion_detect	Constant checking	An object passes by
Bulb_error	Constant checking	A bulb malfunctions
Motion_error	Constant checking	A motion sensor malfunctions
LDR_error	Constant checking	An LDR malfunctions
Connection_error	3 minutes timeout	A node loses connection

Hardware



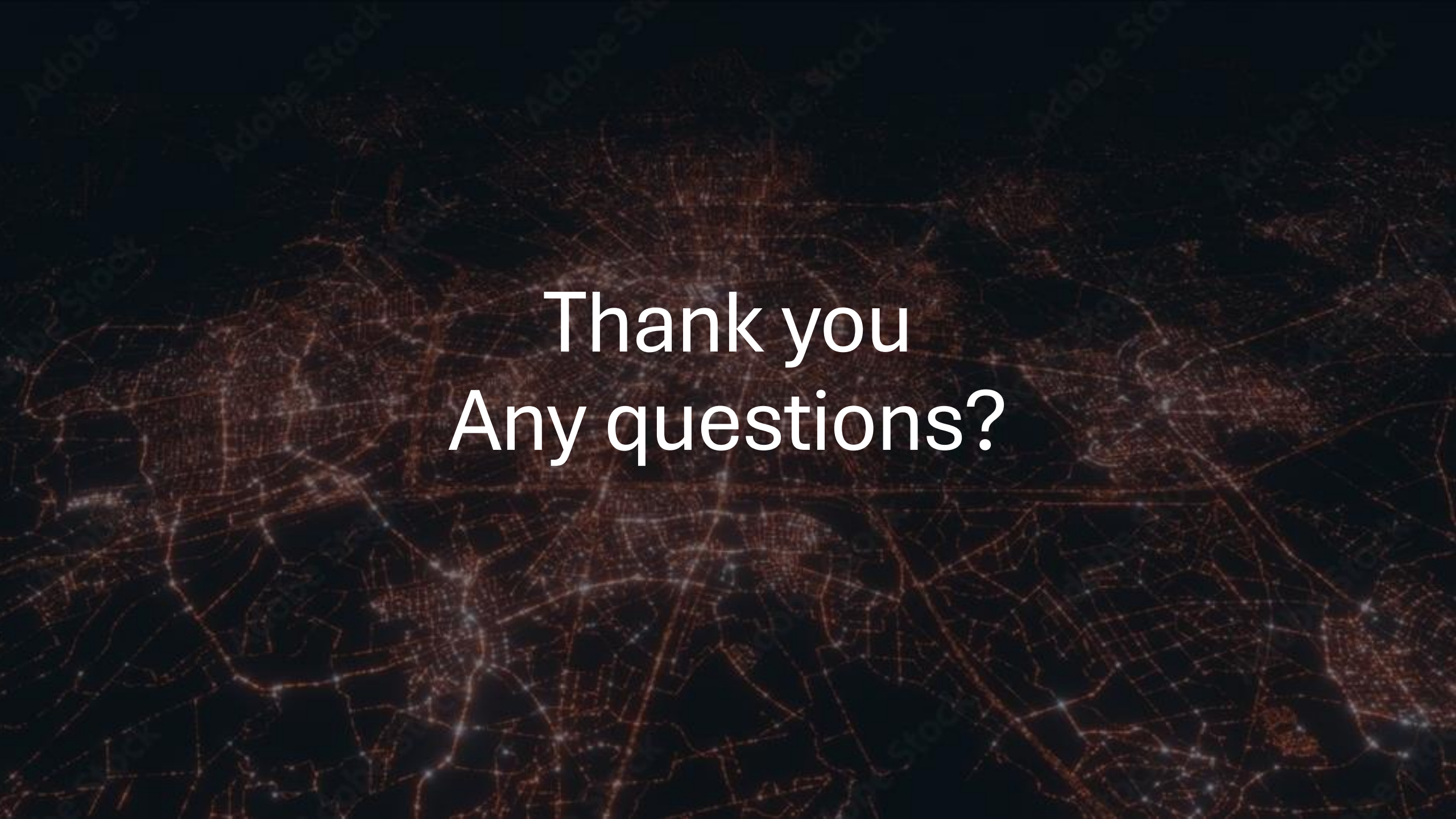
This sensor reads from 2cm to 400cm (0.8inch to 157inch) with an accuracy of 0.3cm (0.1inches)



ESP32 capable of wireless communication



LDR detects external light



Thank you
Any questions?